Fiscal Year:	FY 2023	**Task Last Updated:**	FY 01/08/2023
PI Name:	Karma, Alain Ph.D.		
Project Title:	Computational Studies of Microstructure Formation during Alloy Solidification in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02115-5005	**Congressional District:**	8
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2010 Materials Science NNH10ZTT001N
Start Date:	11/16/2018	**End Date:**	11/15/2023
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 11/15/2023 per NSSC information (Ed., 1/12/23)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC19K0135		
Performance Goal No.:			
Performance Goal Text:			
	NOTE: Continuation of "Integrated Computational and Experimental Studies of Complex Dendritic Microstructure Development during Directional Solidification of Metallic Alloys--NNX16AB54G," with the same Principal Investigator, Dr. Alain Karma. Dendritic microstructures formed during solidification processing (such as casting, welding, and brazing) determine the properties of structural metallic alloys used in aerospace materials essential for NASA exploration in space and produced in numerous US industries with commercial applications ranging from automobile engine parts to turbine blades used in aircraft and power generation. The main project goal is to develop multiscale and multiphysics computational models to relate solidification processing conditions with alloy microstructures and properties. Those models provide enhanced capabilities to accelerate the development of advanced (light-weight, high-temperature, and		

Task Description:	extreme environment) aerospace materials produced on ground by diverse solidification processes including emerging new technologies such as additive manufacturing. In addition, they provide support for materials processing on extraterrestrial surfaces in a reduced gravity environment. Model development is carried out iteratively with microgravity solidification experiments carried out in the Materials Science Laboratory and Transparent Alloy and Device for the Study of Critical Liquids and Crystallization (DECLIC) facilities aboard the International Space Station (ISS) in collaboration with ESA (European Space Agency) and CNES (Centre national d'études spatiales, French space agency) research teams. Computationally-designed materials has been identified as a critical technology in the NASA Technology Roadmap. This research meets the general objective of this critical area of accelerating materials development by integrating physics-based computational models of materials at multiple length scales to relate solidification processing, microstructures, and properties, and by developing massively parallel implementation of computational codes on graphics processing units (computer power). The present project makes use of both phase-field (PF) simulations and the dendritic needle network (DNN) model to establish benchmark comparisons between theory and microgravity experiments. Phase-field simulations resolve the details of the dendrite tip scale, while the DNN model tracks the dynamics of dendritic networks on a much larger scale of several grains. For this reason, the DNN model has emerged as a powerful method to bridge the dendrite tip and grain scales. Those simulation tools provide the means to model microstructure formation in a diffusive growth regime over the full range of scales needed to establish quantitative benchmark comparisons with microgravity observations. Ongoing computational studies have two main objectives: DECLIC/DSI. The first objective is to model directional solidification experiments in the Directional Solidification Insert (DSI) of the DECLIC facility (DEvice for the study of Critical LIquids and Crystallization) onboard the ISS as part of a CNES/NASA project. Those experiments visualize for the first time in dilute transparent alloys the solid-liquid interface evolution in three dimensions under diffusive growth conditions. They provide a wealth of new observations to elucidate complex mechanisms governing the spatiotemporal evolution of three-dimensional dendritic array structures. The goal of this research is to model both DECLIC/DSI experiments and the new series of DECLIC/DSI-reflight(R) experiments. CETSOL (Columnar-to-Equiaxed Transition in Solidification Processing (CETSOL). The second objective is to model directional solidification experiments in metallic and concentrated transparent organic alloys in the framework of the ESA/NASA CETSOL-6 project (March 2018 – February 2021) carried out in collaboration with US and European partners. The general goal of this project is to improve the understanding of complex physical phenomena governing the Columnar-to-Equiaxed Transition (CET) in alloy SOLidification and its consequences for industrial castings. This project involves studies of metallic alloys and the transparent alloy system Neopentylglycol-(d)Camphor (NPG-DC) in the Transparent Alloy facility. This system adds the capability to visualize both columnar and equiaxed dendritic grain development and the CET in a concentrated alloy regime, thereby complementing the DECLIC/DSI-R studies of columnar dendrites in dilute SCN-camphor alloys.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Many commercial structural alloys, from nickel based superalloys used in turbine blades to lightweight aluminum and magnesium alloys used for automotive parts, form dendritic microstructures during the solidification process. These microstructures in turn control the integrity and mechanical properties of cast parts. This research is aimed at developing computational tools to predict reliably the formation of these dendritic microstructures on experimentally relevant length and time scales. Those tools provide enhanced capabilities to accelerate the development of advanced (light-weight, high-temperature, and extreme environment) aerospace materials produced on ground by diverse solidification processes including emerging new technologies such as additive manufacturing. In addition, they provide support for materials processing on extraterrestrial surfaces in a reduced gravity environment.
Task Progress:	The major accomplishments during the present grant period include: • We have performed a detailed quantitative comparison of simulated dendritic grain structures using the dendritic needle network (DNN) method and the corresponding observed grain structures obtained from metallographic analyses of Columnar-to-Equiaxed Transition in SOLidification Processing (CETSOL) B2-FM1 and B2-FM2 Al-7wt%Si alloy samples directionally solidified on board the International Space Station (ISS). Experiments and simulations were used to characterize the type and length of the columnar-to-equiaxed transition (CET) in B2-FM1 as well as the arm spacing of columnar grains in B2-FM2. The simulations show a good agreement with experiments for both samples. For B2-FM1, simulations predict a progressive CET as observed. The length of the transition between the columnar and equiaxed regions and the grain elongation factors are found to be in good agreement with experimental measurements. For B2-FM2 samples without grain refiners, the CET is absent. The dendrite structure remains columnar in both stages I and II and the decrease of the primary dendrite arm spacing with solidification length is well-predicted by the simulations. Those comparisons demonstrate that the DNN method can predict complex dendritic microstructures on experimentally relevant length and time scales. • We have carried out a detailed investigation of grain boundary orientation selection behavior during growth competition between columnar bi-crystals using the dendritic needle network (DNN) method. This study extends our previous phase-field study to the well-developed dendritic regime applicable to the CETSOL experiments. This study enabled quantitative predictions of grain boundary orientation selection over a wide range of temperature gradients and solidification rates. • We have initiated the study of columnar dendritic growth in the transparent Neopentylglycol (NPG) D-Camphor (DC) (NPG-DC) system for <111> crystallographic directions using 3D phase-field (PF) simulations.
Bibliography Type:	Description: (Last Updated: 05/10/2026)
Articles in Other Journals or Periodicals	Song Y, Mota FL, Tourret D, Ji K, Billia B, Trivedi R, Bergeon N, Karma A. "Cell invasion during competitive growth of polycrystalline solidification pattern." ArXiv preprint server. Posted November 15, 2022. https://arxiv.org/abs/2211.08500 , Nov-2022

Articles in Other Journals or Periodicals	Song Y, Tourret D, Karma A. "Scaling laws for dendritic crystal growth in a narrow channel." ArXiv preprint server. Posted November 15, 2022. https://arxiv.org/abs/2211.08309 , Nov-2022
Articles in Peer-reviewed Journals	Song Y, Mota FL, Tourret D, Ji K, Billia B, Trivedi R, Bergeon N, Karma A. "Cell invasion during competitive growth of polycrystalline solidification patterns." Nat Commun. 2023 Apr 19;14(1):2244. https://doi.org/10.1038/s41467-023-37458-0 ; PMID: 37076477; PMCID: PMC10115863 . [Note: previously reported in 2022 as preprint.] , Apr-2023
Articles in Peer-reviewed Journals	Song Y, Tourret D, Karma A. "Scaling laws for two-dimensional dendritic crystal growth in a narrow channel." Physical Review E. 2023 May;107(5):L052801. http://dx.doi.org/10.1103/PhysRevE.107.L052801 . [Note: previously listed in 2022 as preprint. , May-2023
Articles in Peer-reviewed Journals	Ji K, Tabrizi AM, Karma, A. "Isotropic finite-difference approximations for phase-field simulations of polycrystalline alloy solidification." J. Comput. Phys. 2022 May 15;457:111069. https://doi.org/10.1016/j.jcp.2022.111069 , May-2022